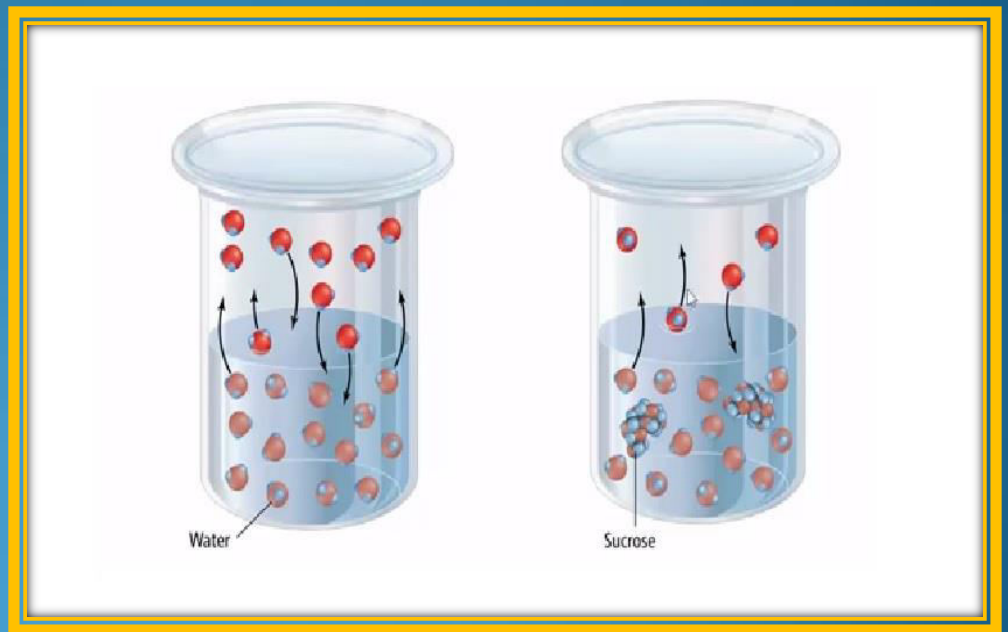


Colligative Properties of Solutions



Physical Properties of Solution

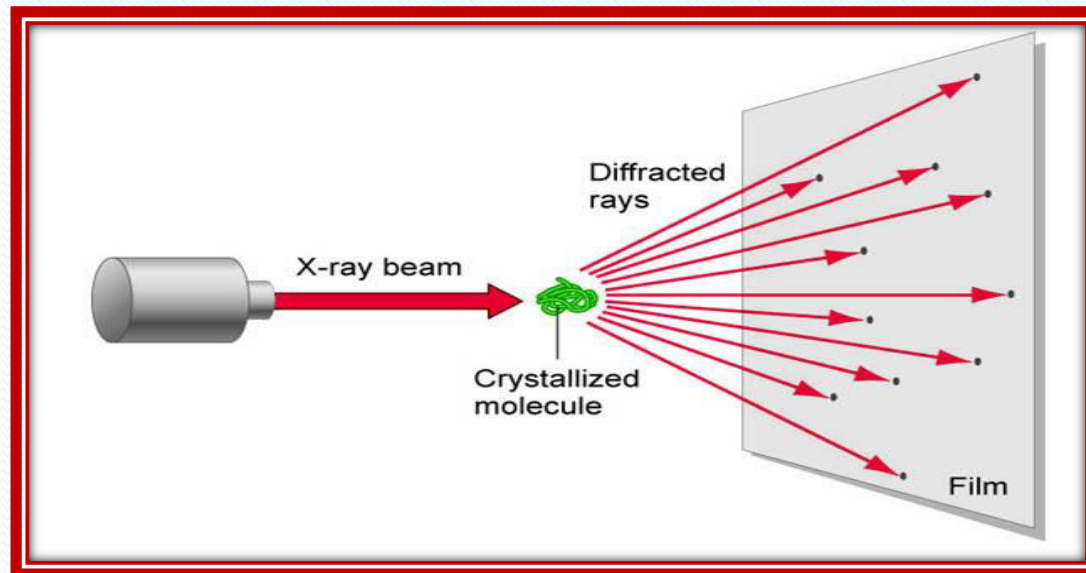
1- Constitutive properties

2- Additive properties

3- Colligative properties

1 - Constitutive properties

- ✓ Properties depend upon → **structural arrangement of atoms** within the molecules.
- ✓ **Ex: Optical properties (Diffraction of light).**
 - **X-ray diffraction:**
 - Diffraction of X-ray beam from atomic planes of crystal → **structure of crystal is investigated.**



2- Additive properties

- ✓ Properties depend on → the **sum of corresponding properties of individual atoms or functional groups** within the molecules.

✓ **Ex:**

1. Molecular weight (M.W.):

- **M.W. = the sum of the atomic weights of the constituent atoms.**

2. Total mass of solution:

- **The sum of masses of the individual components.**

3- Colligative properties

- ✓ Properties depend **ONLY** on → **number of solute particles present** → **NOT** on the **physical or chemical properties of the solute particles**.
- ✓ Solutes may be:
 - 1. **Non-Electrolyte (sucrose).**
 - 2. **Electrolyte (NaCl & Na₂SO₄).**
- ✓ What happens when we put 1 Mole NaCl, Na₂SO₄ & sucrose in 1 Liter water ????
 - Produce **different number of particles**.
 - **1 mole NaCl** → **2 particles** in solvent.
 - **1 mole Na₂SO₄** → **3 particles** in solvent.
 - **1 mole sucrose** → **1 particle** in solvent.
- ✓ **Although** we add same amount of solute (1 mole) → produce **different no. of particles** → **different colligative properties**.

✓ Therefore:

- a) **Colligative properties differ** according to whether solute is **electrolyte or non-electrolyte**.
- b) **Equal concentrations of Non-electrolytes** → **same colligative properties** regardless of chemical nature → **because same number of particles**.
- c) **Equimolar solutions of Electrolytes** will have **HIGHER** colligative properties than **equimolar solutions of Non-electrolytes**.

Colligative Properties of Solutions

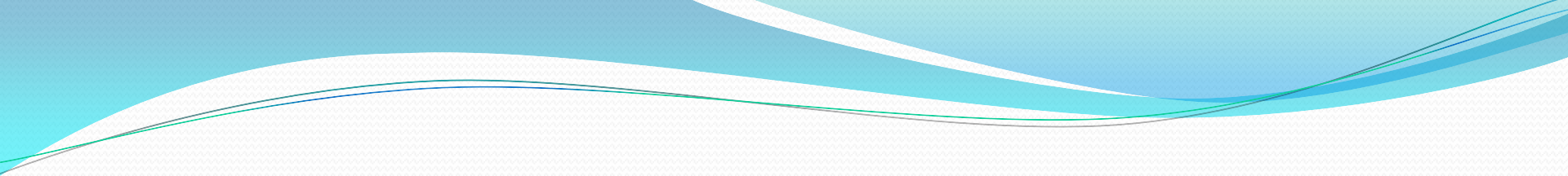
1- Vapor Pressure (V.P.) Lowering

2- Boiling Point (B.P.) Elevation

3- Freeze Point (F.P.) Depression

4- Osmotic Pressure (O.P.)

- ✓ All colligative properties are **inter-related to each other**.
- ✓ **So, for any solution** → **the amount of the vapor pressure lowering, boiling point elevation or freeze point depression** with respect to pure solvent → **depends on the number of solute particles in solution.**



1- Vapor Pressure (V.P.) Lowering

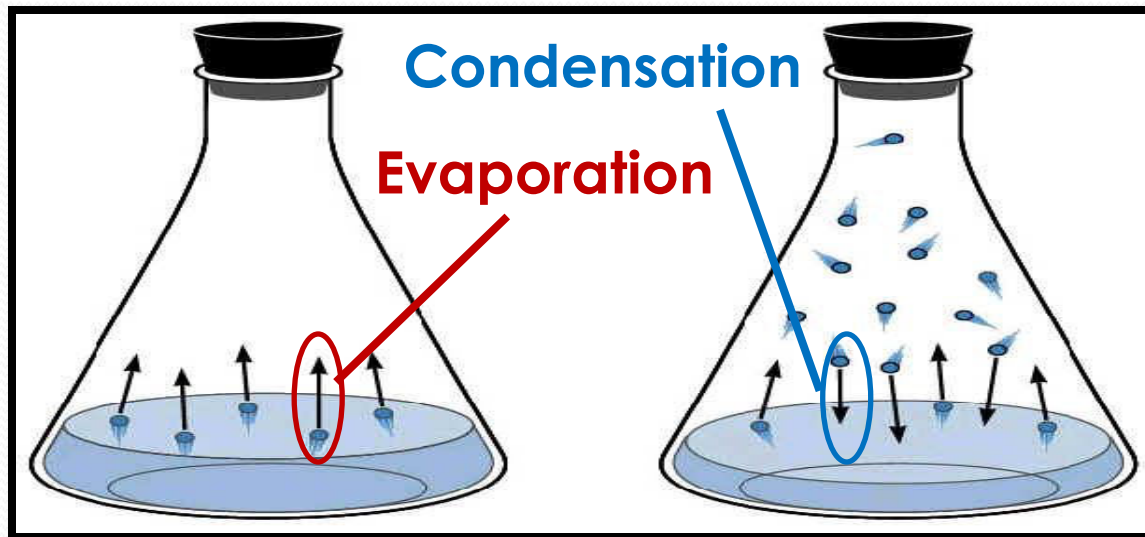
Vapor Pressure (V.P.)

✓ **Definition:**

Pressure exerted by a vapor in dynamic equilibrium with its condensed phases at a given temperature in a **closed system**.

✓ **Equilibrium vapor pressure indicates:**

- i- **Liquid's evaporation rate.**
- ii- The **escaping tendency of particles** from liquid.



✓ **Volatile substance** → substance with **high vapor pressure** at normal temperature.

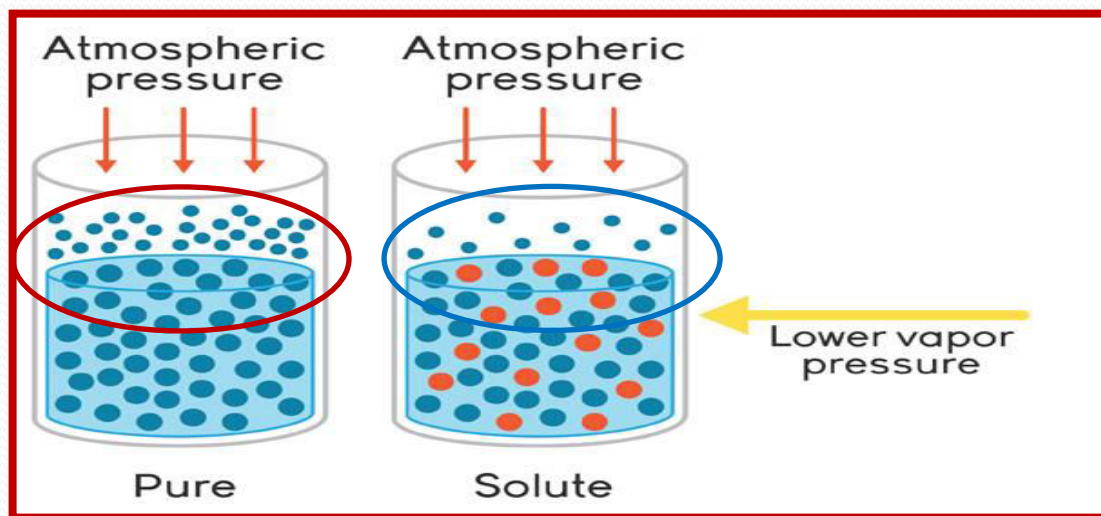
➤ **Vapor Pressure ↑ → Volatility ↑.**

➤ **Vapor Pressure ↓ → Volatility ↓.**

Vapor Pressure Lowering

What if we add **NON-VOLATILE** solute to this solvent

- ✓ **Solvent molecules on the surface** (normally escape into vapor and contribute to the vapor pressure) are **replaced by solute molecules**.
- ✓ **The solute is non-volatile** → **does not contribute** to the vapor pressure → **lowering the vapor pressure of the solution**.



Why is V.P. Lowering a colligative property???

- ✓ Because $\uparrow$ **number of solute particles** in solution $\rightarrow \uparrow$ **particles that replace solvent molecules on surface** $\rightarrow \downarrow$ **escaping solvent molecules** $\rightarrow$ **Lowers V.P. of solution.**

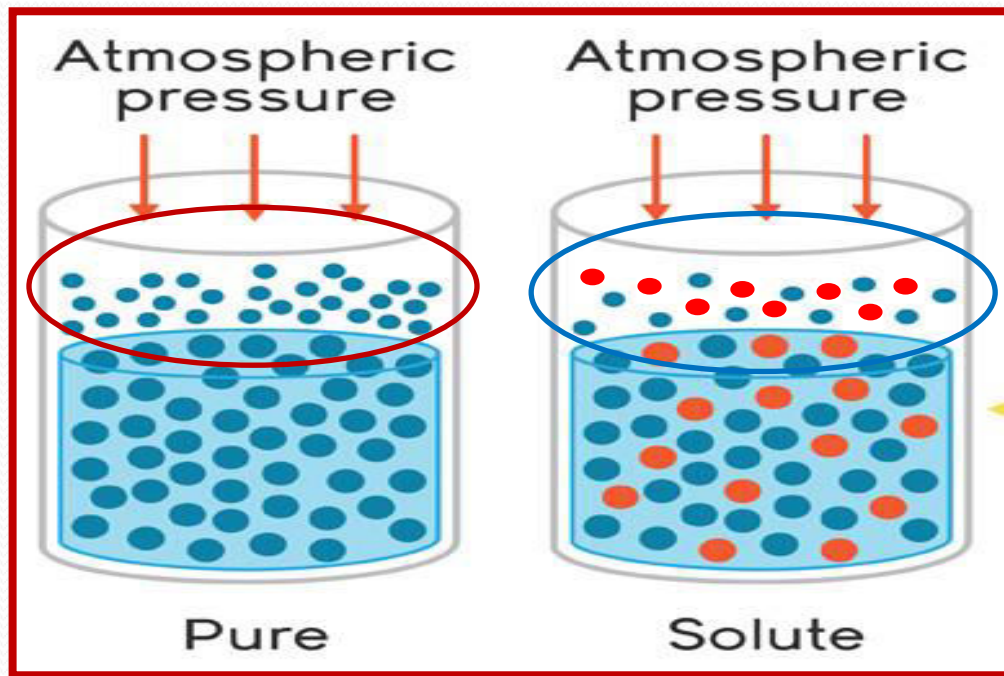
Conclusion

$\therefore$ When a non-volatile solute is dissolved in a pure solvent:

- 1) **V.P. of solution $<$ pure solvent.**
- 2) **V.P. lowering** $\rightarrow$ **directly proportional** to the **number of solute molecules or ions** $\rightarrow$ **colligative property.**

What if we add **VOLATILE** solute to this solution

- ✓ **Volatile solute** will volatilize → **contribute to the total vapor pressure** above the solution.



Raoult's Law of Vapor Pressure

- ✓ The vapour pressure of an **ideal solution** is directly dependent on:
 - i) The **vapour pressure of each chemical component** (solute + solvent).
 - ii) The **mole fraction of each components** present in the solution.

- ✓ To calculate the **Total V.P. above the solution (P_{total})** → we must calculate:
 - 1) **Partial V.P. exerted by the solvent (P_{solvent}).**
 - 2) **Partial V.P. exerted by the volatile solute ($P_{\text{volatile solute}}$).**
- ✓ **The Sum** of the two → **Total V.P.** of the solution.

$$\therefore \text{Raoult's Law} \rightarrow P_{\text{total}} = P_{\text{solvent}} + P_{\text{volatile solute}}$$

✓ $P_{\text{total}} \rightarrow$ **Total V.P.**

✓ $P_{\text{solvent}} = P_{\text{solvent}}^{\circ} X_{\text{solvent}}$

➤ P_{solvent} $\rightarrow$ **Partial V.P. of solvent** in solution.

➤ $P_{\text{solvent}}^{\circ}$ $\rightarrow$ **V.P. of pure solvent.**

➤ X_{solvent} $\rightarrow$ **mole fraction of solvent.**

✓ $P_{\text{volatile solute}} = P_{\text{volatile solute}}^{\circ} X_{\text{volatile solute}}$

➤ $P_{\text{volatile solute}}$ $\rightarrow$ **Partial V.P. of volatile solute** in solution.

➤ $P_{\text{volatile solute}}^{\circ}$ $\rightarrow$ **V.P. of pure volatile solute.**

➤ $X_{\text{volatile solute}}$ $\rightarrow$ **mole fraction of volatile solute.**

$$\therefore P_{\text{total}} = [P_{\text{solvent}}^{\circ} X_{\text{solvent}}] + [P_{\text{volatile solute}}^{\circ} X_{\text{volatile solute}}]$$

✓ **Example:**

What is the **total vapor pressure** of an ideal solution of benzene and toluene in which the **mole fraction of benzene is 0.6**? (Vapor pressure of pure benzene is 95.1 mm Hg and the vapor pressure of pure toluene 28.4 mm Hg at 25°C).

$$\therefore P_{\text{total}} = P_{\text{benzene}} + P_{\text{toluene}}$$

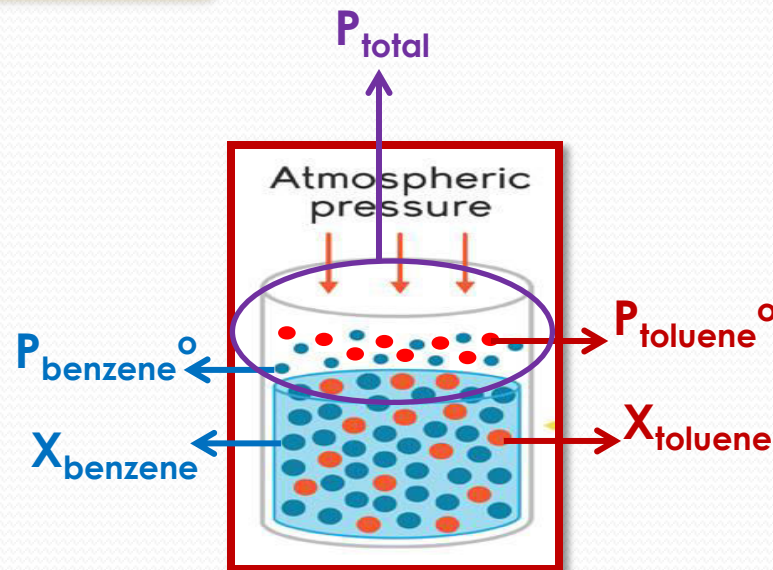
✓ **Mole fraction of benzene** = 0.6.

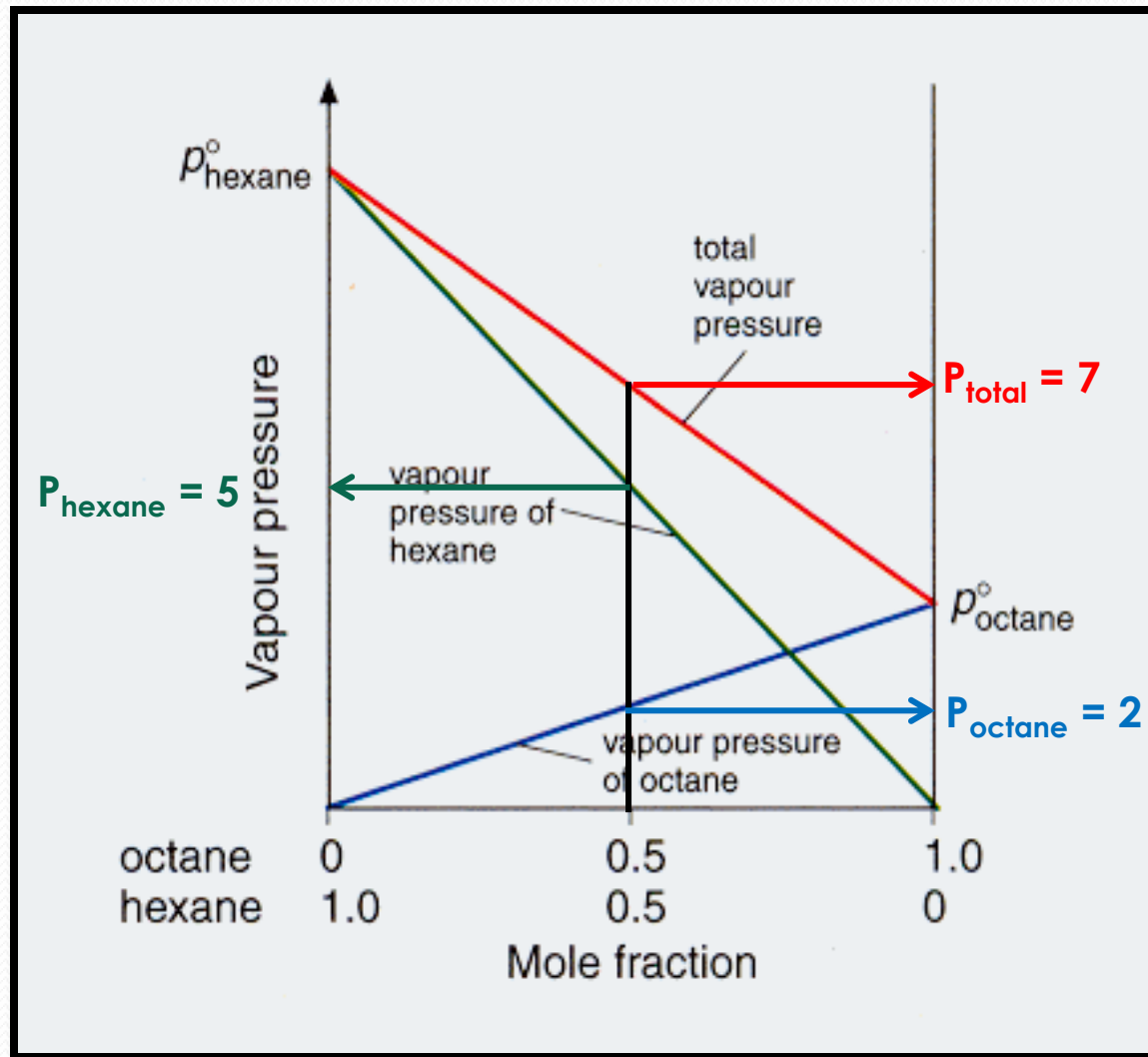
$\therefore$ **Mole fraction of toluene** = $1 - 0.6$
= **0.4**.

✓ $P_{\text{benzene}} = P_{\text{benzene}}^{\circ} X_{\text{benzene}}$
= $95.1 \times 0.6 = \mathbf{57.1 \text{ mm Hg}}$.

✓ $P_{\text{toluene}} = P_{\text{toluene}}^{\circ} X_{\text{toluene}}$
= $28.4 \times 0.4 = \mathbf{11.4 \text{ mm Hg}}$.

$\therefore P_{\text{total}} = P_{\text{benzene}} + P_{\text{toluene}}$
= $57.1 + 11.4 = \mathbf{68.5 \text{ mm Hg}}$.





Raoult's Law of V.P. for Non-Volatile solute (ex: Non-electrolyte)

- ✓ The solute → **does not contribute** to the V.P. → **lowering the vapor pressure of the solution.**

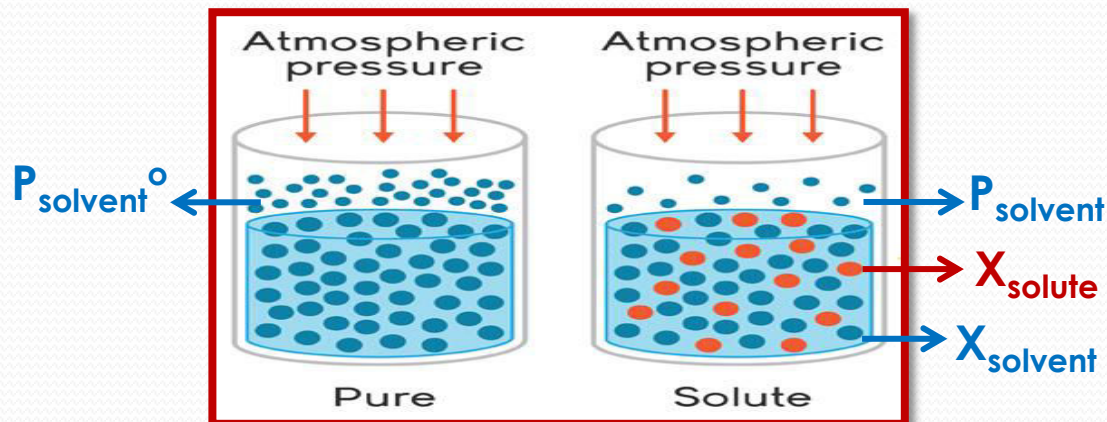
$$\therefore P_{\text{total}} = P_{\text{solvent}}$$

$$\therefore P_{\text{solvent}} = P_{\text{solvent}}^{\circ} X_{\text{solvent}}$$

- ✓ How to calculate V.P. Lowering (ΔP)??

$$\therefore \Delta P = P_{\text{solvent}}^{\circ} - P_{\text{solvent}}$$

$$\therefore \Delta P = P_{\text{solvent}}^{\circ} X_{\text{solute}}$$



✓ **Example:**

Calculate the **V.P. lowering** when **175 g of sucrose (MW = 342.30)** is dissolved into **350 ml water (MW = 18.02)** at 75°C. The **V.P. of pure water** at 75°C is **289.1 mm Hg**.

$$\therefore \Delta P = P_{\text{water}}^{\circ} - P_{\text{water}}$$

$$\therefore P_{\text{water}} = P_{\text{water}}^{\circ} X_{\text{water}}$$

- ✓ **No. of moles of sucrose = weight/MW = 175/342.3 = 0.511 moles.**
- ✓ **No. of moles of water = weight/MW = 350/18.02 = 19.42 moles.**
- ✓ **Total no. of moles = 0.511 + 19.42 = 19.934 moles.**
- ✓ **X_{sucrose} = 0.511/19.934 = 0.0256.**
- ✓ **X_{water} = 19.42/19.934 = 0.974.**

$$\therefore P_{\text{water}} = P_{\text{water}}^{\circ} X_{\text{water}} = 289.1 \times 0.974 = \mathbf{281.64 \text{ mm Hg.}}$$

$$\therefore \Delta P = P_{\text{water}}^{\circ} - P_{\text{water}} = 289.1 - 281.64 = \mathbf{7.4 \text{ mm Hg.}}$$

OR

$$\therefore \Delta P = P_{\text{water}}^{\circ} X_{\text{sucrose}} = 289.1 \times 0.0256 = \mathbf{7.4 \text{ mm Hg.}}$$

Practical application of V.P. Lowering

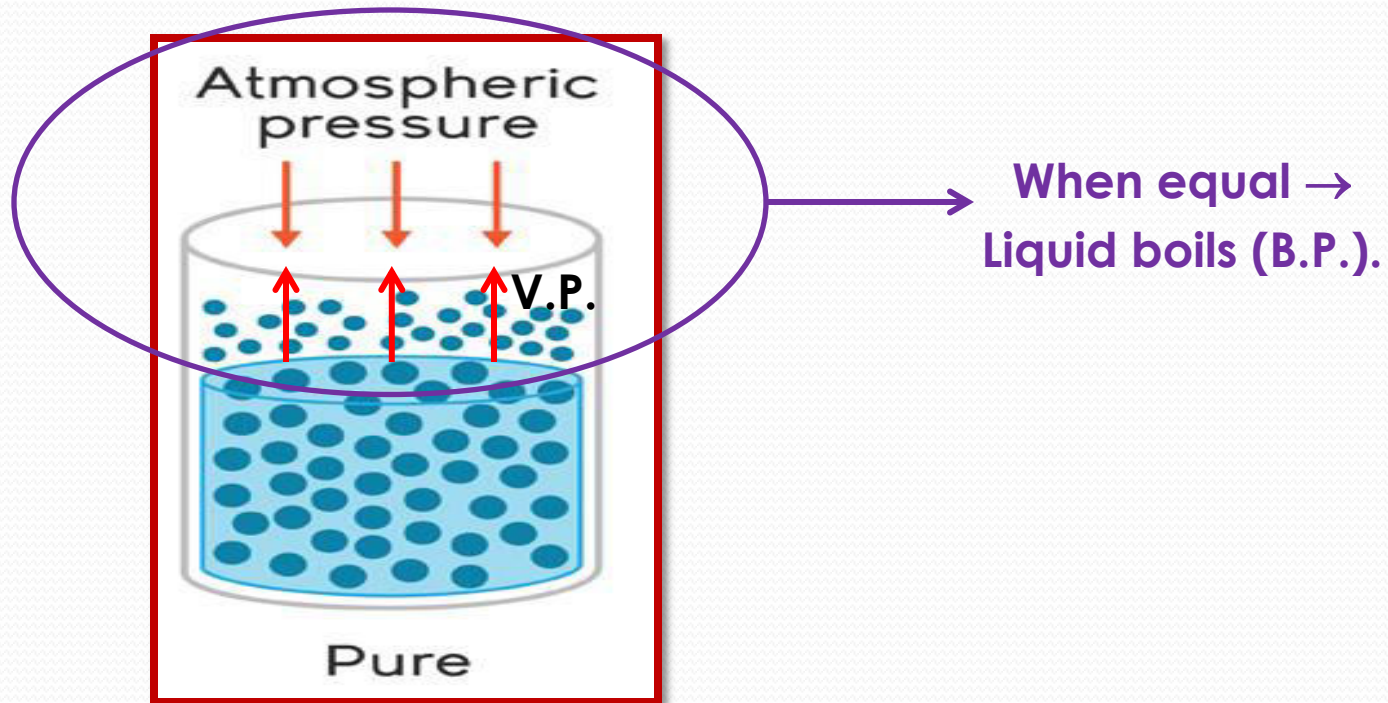
- ✓ **Calculate the V.P. of aerosol propellant** → as mixture of liquefied gas propellants is considered a solution.

2- Boiling Point (B.P.) Elevation

Boiling Point (B.P.) of a liquid

✓ **Definition:**

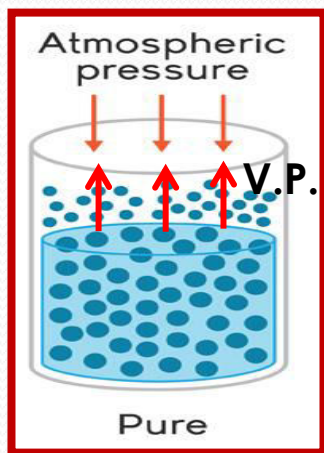
The temperature at which the **vapor pressure of the liquid** is equal to the **external atmospheric pressure** (760 mm Hg).



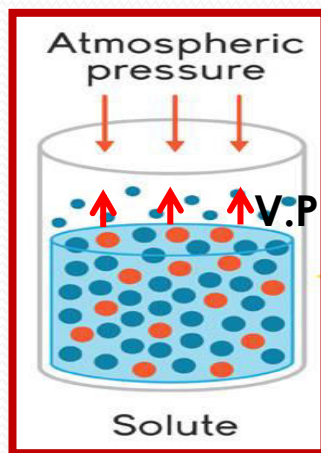
Boiling Point Elevation

What if we add **NON-VOLATILE** solute to this liquid

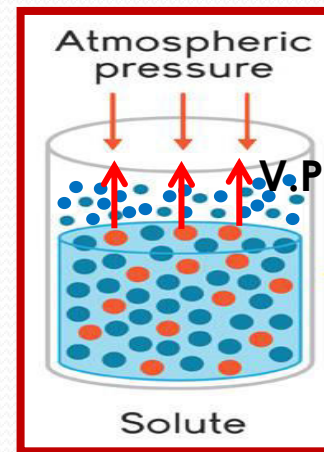
✓ ↓ **V.P. of solution** → solution is **heated to higher temperature** → ↑ **kinetic energy** → **reach same V.P.** (equal to atmospheric) & boils → **B.P. Elevation**.



B.P. = 100°C



100°C → No Boiling



B.P. = 120°C

∴ 1) **B.P. of solution >> pure solvent.**

2) **B.P. elevation** → directly proportional to the **number of solute molecules (mole fraction)** → **colligative property**.

$$\Delta T_b = K_b m$$

- ✓ B.P. elevation is **proportional to the molality** of the solution:

$$\Delta T_b = K_b m$$

- ΔT_b → B.P. Elevation.
- m → molality of solution.
 - **Molality** → no. of moles of solute/ kilograms of solvent.
- K_b → the **molal elevation constant** or the **ebullioscopic constant**.
 - **Def.:** the **B.P. elevation for one molal** dilute solution.
 - **Unit:** °C/molal.
 - **Characteristic property for each solvent** → for water it is **0.515** °C/molal.

Practical application of B.P. Elevation

1) Calculate solute conc. in solution from previous equation:

- By measuring the B.P. elevation of solution and knowing K_b of solvent.

2) Calculate the MW of solute.

✓ **Example:**

A solution of dextrose (MW = 180.2) in 1000 grams of water has a boiling point of 104.3 °C at a pressure of 1 atm. How many grams of dextrose are there in the solution? (K_b water = 0.52 °C/molal).

$$\Delta T_b = K_b m$$

✓ $\Delta T_b = 104.3 - 100 = 4.3$ °C.

✓ $\Delta T_b = K_b m$.

∴ $4.3 = 0.52 m$.

∴ $m = 8.27$.

✓ **Molality** = no. of moles/ wt. solvent in kg.

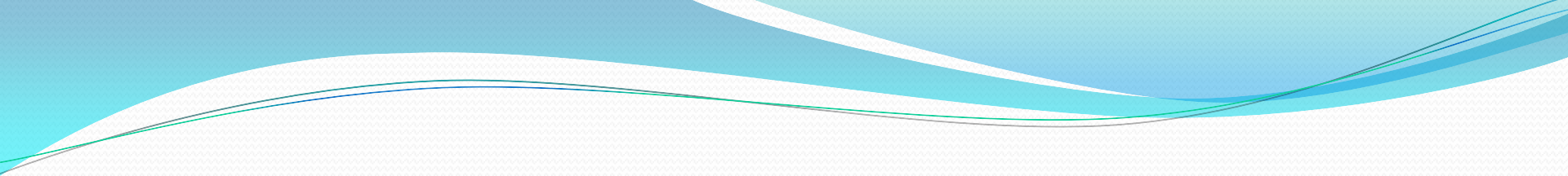
∴ $8.27 = \text{no. of moles} / 1 \text{ kg}$.

∴ **No. of moles of dextrose** = 8.27 moles.

✓ **No. of moles of dextrose** = weight/MW.

∴ $8.27 = \text{weight} / 180.2$.

∴ **Weight of dextrose** = 1490.25 g.

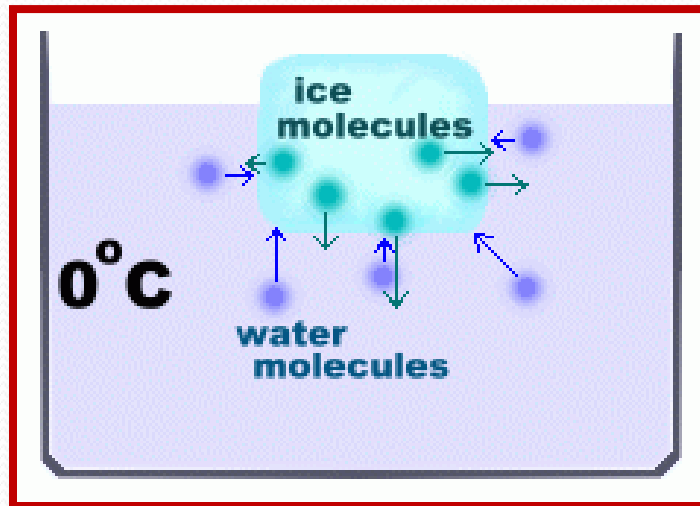


3- Freezing Point (F.P.) Depression

Freezing Point (F.P.) of a liquid

✓ **Definition:**

The temperature at which the **solid and liquid phases are in equilibrium** under a pressure of **1 atmosphere (760 mm Hg)**.



Freezing Point Depression

What if we add NON-VOLATILE solute to this liquid

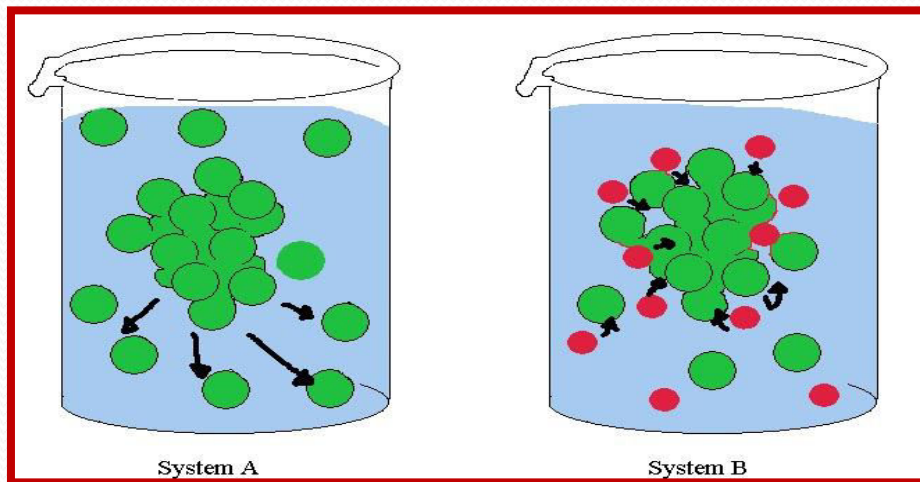
✓ Solute $\rightarrow$ $\downarrow$ F.P. of solution than pure solvent $\rightarrow$ F.P. Depression.

$\therefore$ 1) F.P. of solution $\ll$ pure solvent.

2) F.P. depression $\rightarrow$ directly proportional to the number of solute molecules (mole fraction) $\rightarrow$ colligative property.

Why F.P. depression takes place???

- a) **For a liquid to freeze** → **molecules must arrange in a cluster** → grows into solid.
- b) **When solute is added to solvent** → **↑ entropy** (disordered arrangement of solvent molecules).
- c) **Solute particles** → **hinder solvent molecules from coming together** to form the cluster.
- d) **To reach the solid cluster** → **solvent molecules must be slowed down**.
- e) **∴ ↓ Temp.** → **to ↓ kinetic energy** → Freezing → **F.P. Depression**.



$$\Delta T_f = K_f m$$

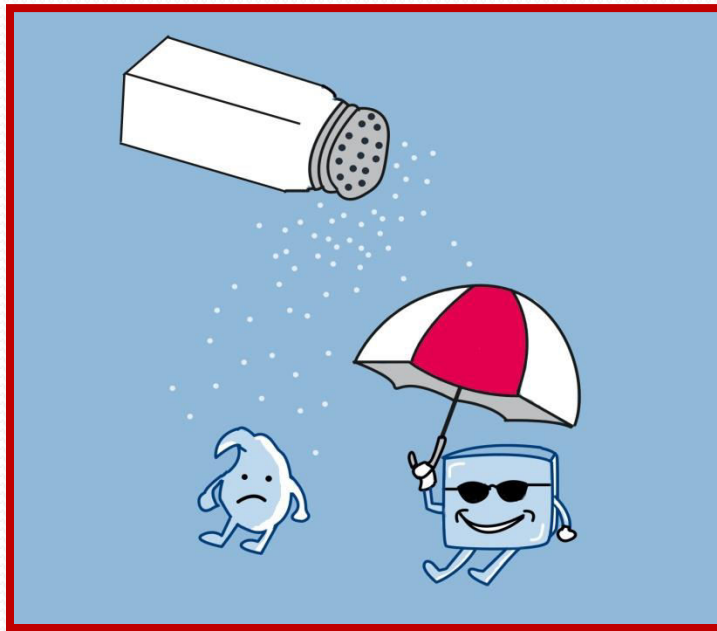
✓ F.P. depression is **proportional to the molality** of the solution:

$$\Delta T_f = K_f m$$

- ΔT_f → F.P. Depression.
- m → **molality** of solution.
- K_f → the **molal F.P. depression constant** or the **cryoscopic constant**.
 - **Def.:** the **F.P. depression for one molal** dilute solution.
 - **Unit:** **°C/molal**.
 - **Characteristic property for each solvent** → for water it is **1.86** °C/molal.

Practical application of F.P. Depression

- 1) **Salt addition to wet cold roads → prevent them from becoming icy** when temp. drops below zero.
- 2) **Calculate Freezing point of a solution.**
- 3) **Calculate solute conc. in solution.**
- 4) **Calculate the MW of solute.**



✓ **Example:**

If 52.5 g of LiF (MW = 25.94) is dissolved in 1000 g of water, what is the expected freezing point of the solution? (K_f water = 1.86°C/molal)

$$\Delta T_f = K_f m$$

✓ **Molality = no. of moles/ wt. solvent in kg.**

✓ **No. of moles of LiF = weight/MW.**

∴ **No. of moles of LiF** = $52.5 / 25.94 = 2.02$ moles.

∴ **m = no. of moles/ wt. solvent in kg.**

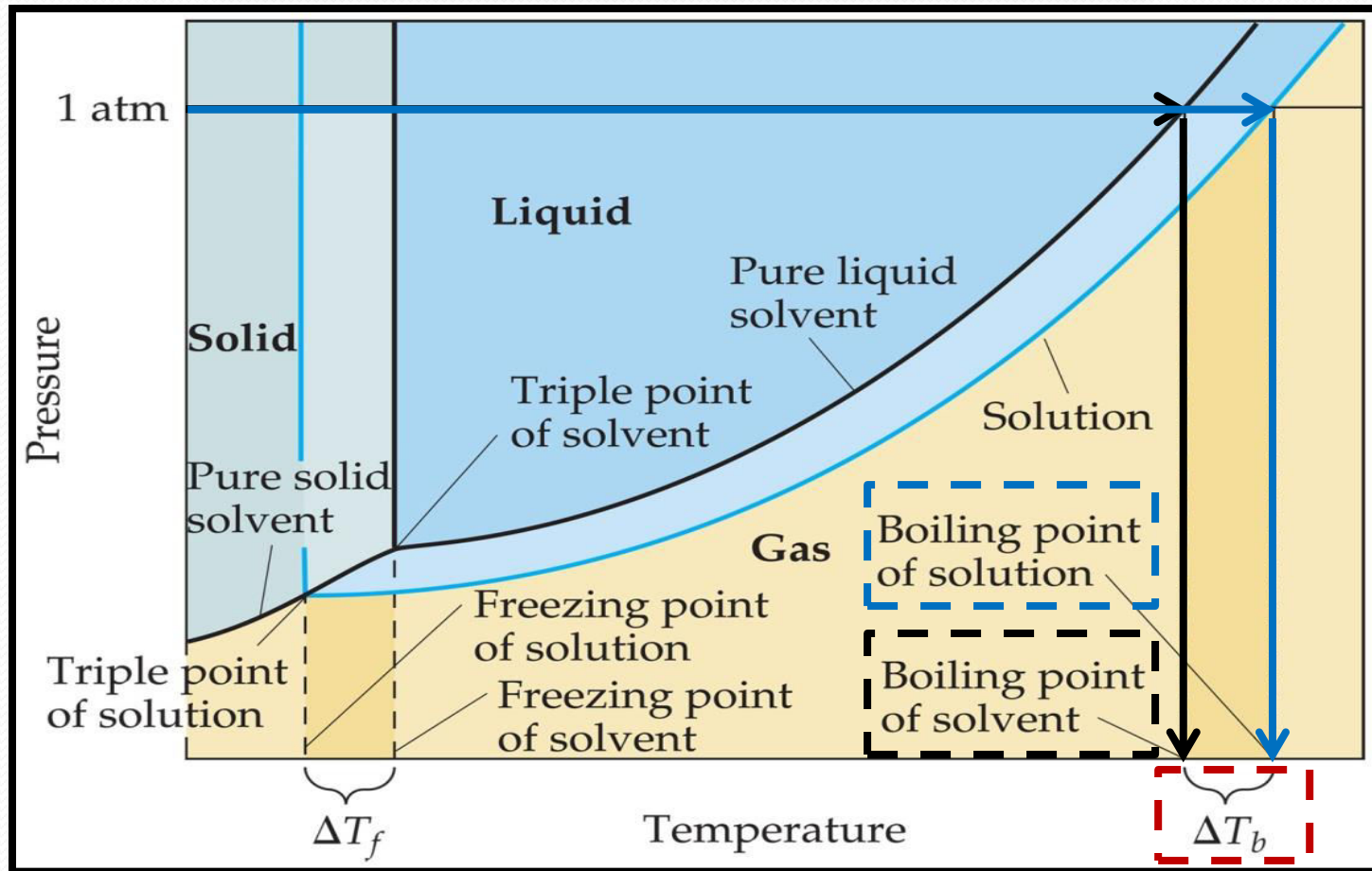
∴ **m** = $2.02 / 1 \text{ Kg} = 2.02$ molal.

✓ **$\Delta T_f = K_f m$.**

∴ **ΔT_f** = $1.86 \times 2.02 = 3.76^\circ\text{C}$.

∴ **F.P. of solution** = $0 - 3.76 = - 3.76^\circ\text{C}$.

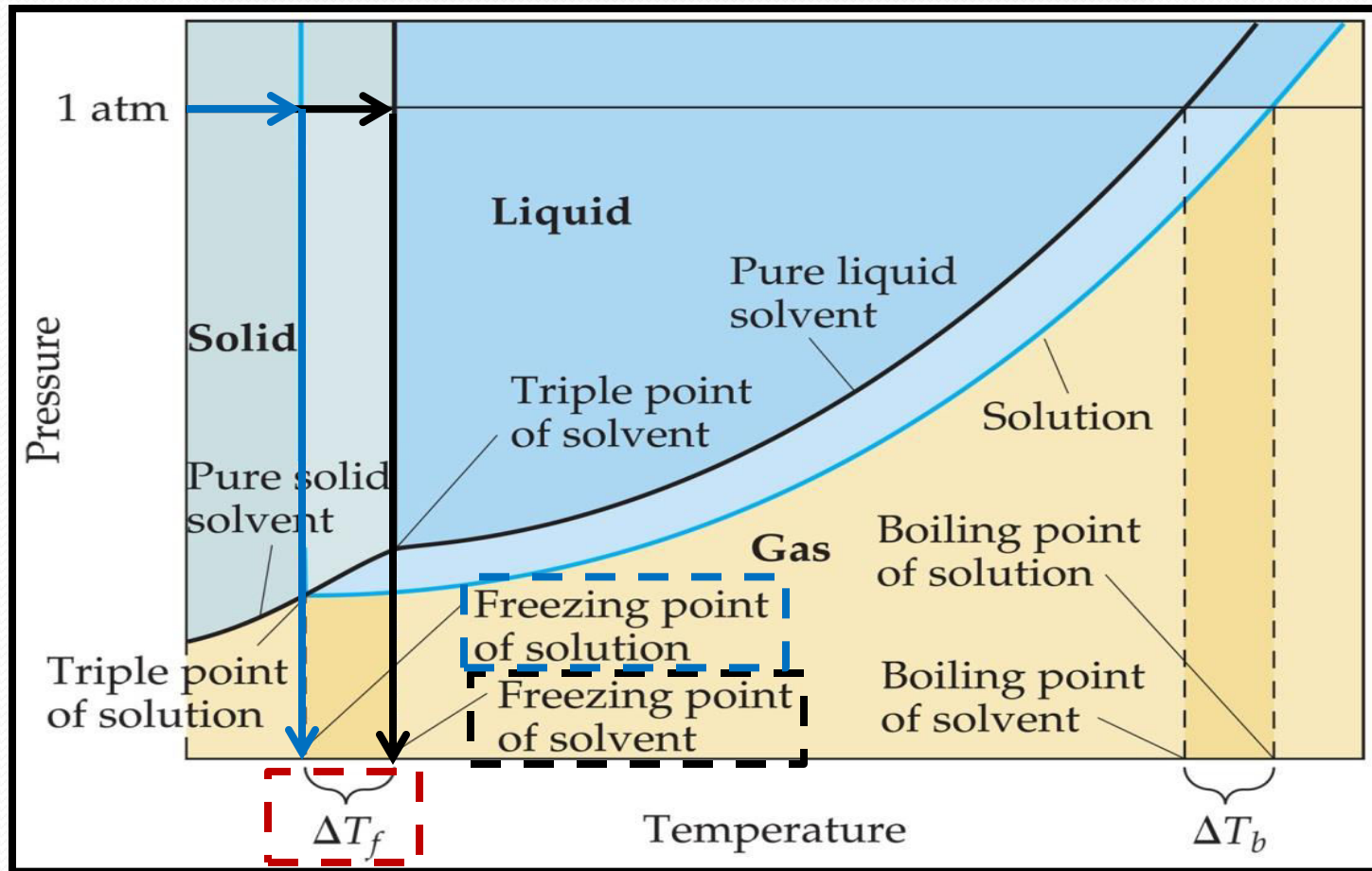
Phase diagram of pure water



✓ In Solution (blue curve):

- Boundary line between liquid & gas is **shifted to the right** → **Lower V.P. → B.P. Elevation.**

Phase diagram of pure water



✓ In Solution (blue curve):

- Boundary line between liquid & solid is **shifted to the left** → **F.P. Depression.**

Remember

V.P. Lowering

B.P. Elevation

F.P. Depression

✓ They are **COLLIGATIVE** properties.

✓ They depend on the **NUMBER of PARTICLES OR MOLE FRACTION OR MOLAL CONC. of SOLUTE.**

✓ The **MORE concentrated** the solution → the **GREATER** the V.P. Lowering, B.P. Elevation & F.P. Depression.

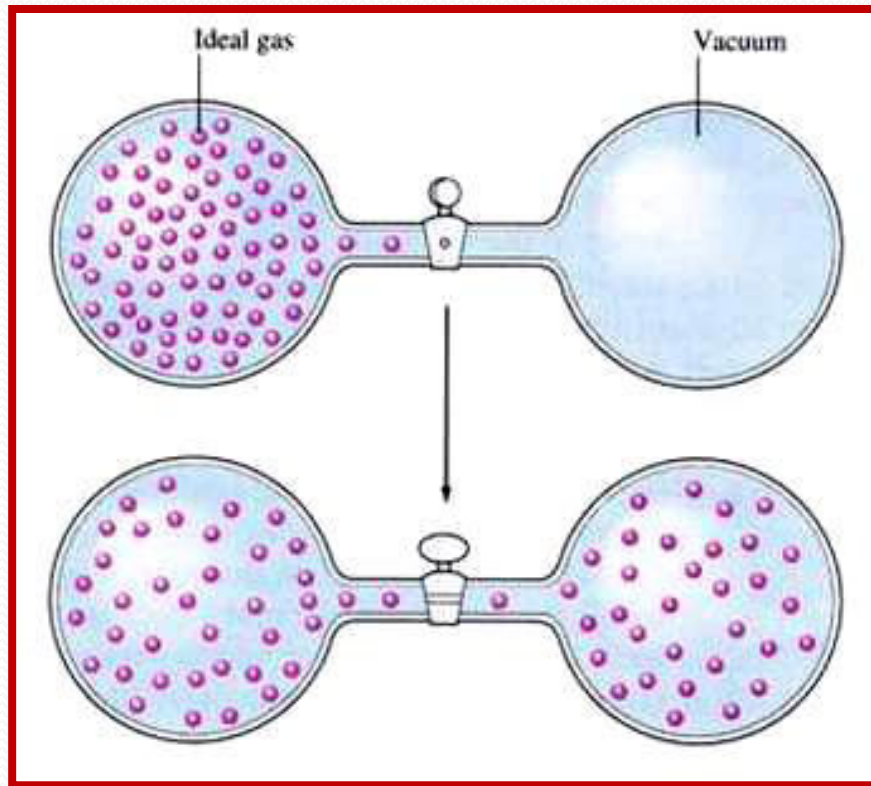
4- Osmotic Pressure (O.P.)

Osmotic Pressure (O.P.)

- ✓ The **most important** of colligative properties → because **related to the physiological compatibility of parenteral, ophthalmic and nasal solutions** with body fluids.
- ✓ **Difficult to measure.**
- ✓ So, other **colligative properties (especially F.P. depression) are determined** → then **related to the O.P.** of solution.

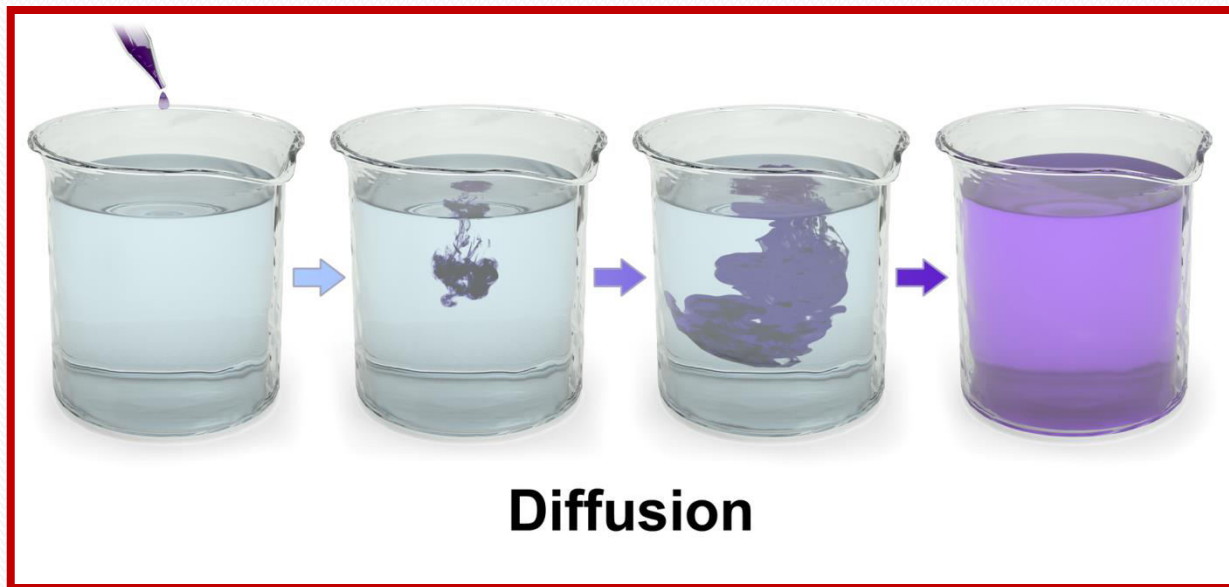
Diffusion in Gases

- ✓ Gases diffuse **from area of high gas concentration** → **to area of lower gas concentration**.



Diffusion in Liquids

- ✓ **Molecules or ions** in liquids have the same **freedom to move** as gases.
- ✓ **Ex:**
 - Put free drops of a **dye in water**.
 - **Dye start to move** into the water.
 - Water start to migrate from layer to layer.
 - At last → **the composition of the new solution is uniform**.

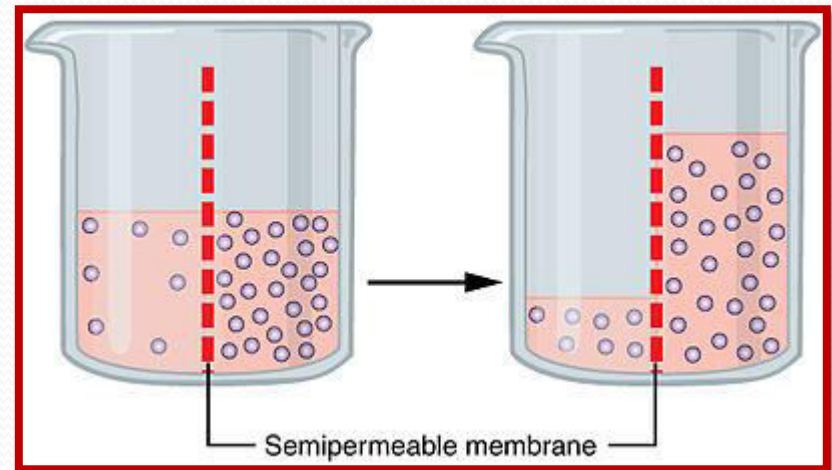
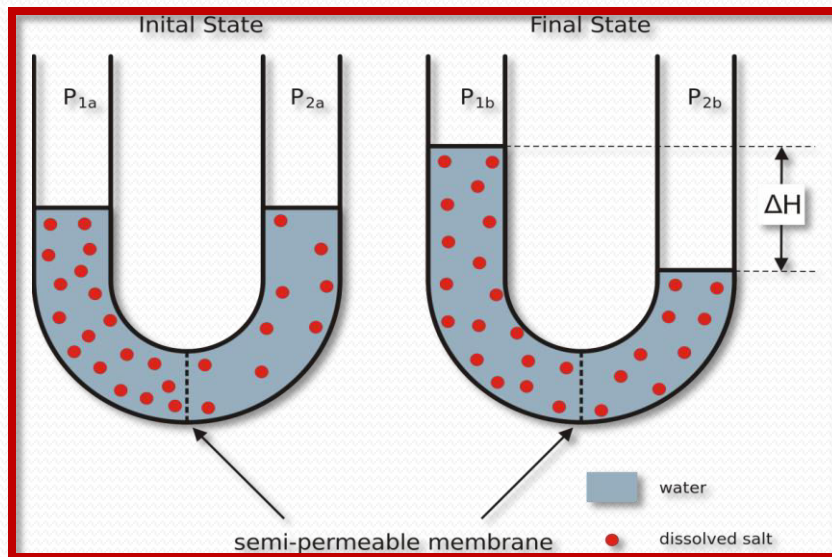


✓ **So, in Diffusion:**

- Substances diffuse from **regions of higher concentrations** to the **regions of lower concentrations** → so that the difference in conc. eventually disappeared.

Osmosis

- ✓ Consider a pure solvent and a salt solution separated by a semipermeable membrane:
 - **Solvent molecules** move through the membrane.
 - Solvent molecules move from **pure solvent (high solvent)** into the **solution (low solvent)**.
 - This movement continues till **equilibrium** → solvent conc. is equal on both membrane sides.



∴ Osmosis

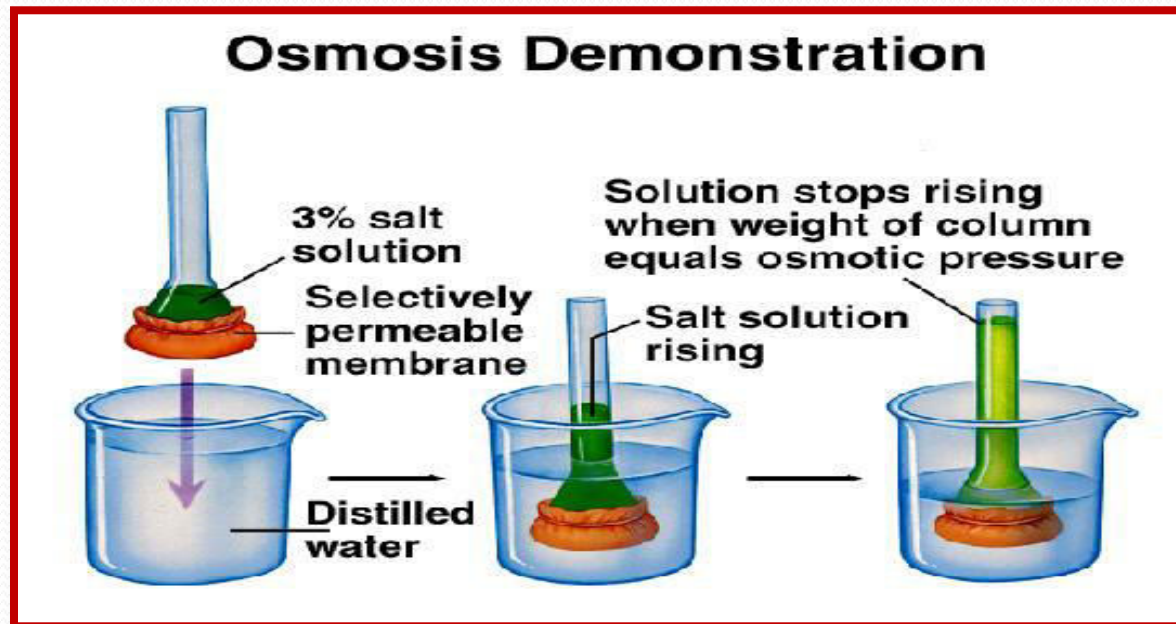
- ✓ Diffusion of solvent molecules through a semipermeable membrane from **solution of lower solute concentration (i.e: diluted → high solvent)** to **solution of higher solute concentration (i.e: concentrated → low solvent)**.

Osmotic Pressure (O.P.)

✓ In the following experiment :

- Water molecules will **move from the beaker to inside the tube** (osmosis).
- **Solution height will rise inside the tube until:**

$$\begin{array}{ccc} \text{Hydrostatic pressure} & = & \text{Osmotic pressure} \\ \text{(exerted by solution column)} & & \text{(Due to escaping tendency)} \end{array}$$



$\therefore$ Osmotic Pressure (O.P.)

- ✓ The **pressure that must be applied to the solution to prevent the migration of pure solvent molecule** from the more diluted (pure solvent) side into the solution.

Osmotic Pressure (O.P.) is a colligative property

- ✓ Because, O.P. of a solution is dependent only on the number of solute particles, not their nature.

N.B: Cell membrane $\rightarrow$ is a semipermeable membrane.

Osmotic Pressure of Non-electrolytes

- ✓ Two non-electrolyte solutions with same molal concentration $\rightarrow \therefore$ same number of particles $\rightarrow \therefore$ **equal colligative properties**.
 - $\therefore$ **Identical osmotic pressure** $\rightarrow$ the 2 solutions are **ISO-Osmotic**.
 - **Ex:** 0.1 molal sucrose solution is **Iso-Osmotic** with 0.1 molal dextrose solution.
- ✓ O.P. is proportional to the conc. of non-electrolyte.
 - **Two fold increase** in conc. of non-electrolyte $\rightarrow$ **doubling of O.P.**
 - **Ex:**
 - **0.1 molal** dextrose $\rightarrow$ O.P. = **2.4 atm.**
 - **0.2 molal** dextrose $\rightarrow$ O.P. = **4.8 atm.**
 - **1 molal** dextrose $\rightarrow$ O.P. = **24 atm.**

Van't Hoff Equation for O.P.

- ✓ In dilute solutions, the **O.P. (π) is directly proportional to the molarity of the solution (mole/liter) and to its temperature ($^{\circ}\text{K}$).**

$$\pi = MRT$$

- $\pi \rightarrow$ **O.P. in atm.**
- $M \rightarrow$ **molarity** of solution (**no. of moles/ 1L of solution**).
 $\rightarrow$ **molarity = no. of moles / volume (in L).**
- $R \rightarrow$ **ideal gas constant = 0.082 L atm./ mole deg.**
- $T \rightarrow$ **absolute temperature ($^{\circ}\text{K} = ^{\circ}\text{C} + 273$).**

- ✓ Since, $M = \text{no. of moles/ volume (L)} = n/V$.

- $\therefore$ Van't Hoff equation is:

$$\pi = \frac{n}{V} RT$$

✓ **Example:**

Calculate the osmotic pressure of a solution containing 1g of sucrose (MW = 342) dissolved in 100 ml water at 20°C. (R = 0.082 L atm./ mole deg.)

$$\pi = \frac{n}{V} RT$$

✓ **n** = weight/MW.

∴ **n** = 1 / 342 = **0.0029 moles.**

✓ **V** = 100 ml = **0.1 L.**

✓ **T** = 20 + 273 = **293 °K.**

$$\pi = \frac{n}{V} RT$$

$$\pi = \frac{0.0029}{0.1} 0.082 \times 298 = 0.71 \text{ atm.}$$

Osmotic Pressure of Electrolytes

✓ Van't Hoff stated that → O.P. of solutions of Electrolytes (ionized substances as strong acids, bases & salts) >>> O.P. of solutions of Non-Electrolytes.

✓ This is mainly due to:

➤ Electrolytes ionize into +ve & -ve charged ions → ↑ number of particles in solution → ↑ O.P.

✓ ∴ Van't Hoff equation for electrolytes become:

$$\pi = i M R T$$

$$\pi = i \frac{n}{V} R T$$

➤ i → number of ions produced by electrolyte ionization.

i

✓ Strong electrolytes → complete ionization.

➤ ∴

i = numbers of ions produced from dissociation of the electrolyte.

✓ Weak electrolytes → incomplete ionization.

➤ ∴

total number of particles (ions & molecules) in the solution

i = $\frac{\text{total number of particles (ions \& molecules) in the solution}}{\text{Number of particles that would be present if the solute did not ionized}}$

Note

✓ This rule is also valid for B.P. elevation & F.P. depression of electrolytes:

$$\Delta T_b = i K_b m \quad \& \quad \Delta T_f = i K_f m$$

✓ **Example:**

Calculate the osmotic pressure exhibited by a 0.42 M KOH solution at 30°C. (R = 0.082 L atm./ mole deg.)



$$\pi = i MRT$$

✓ $i = 2.$

✓ $T = 30 + 273 = 303 \text{ }^\circ\text{K}.$

✓ $\pi = i MRT$

$$= 2 \times 0.42 \times 0.082 \times 303 = \mathbf{20.87 \text{ atm.}}$$

✓ **Example:**

0.5 g hemoglobin was dissolved in water to make 100 ml solution. At 25°C the osmotic pressure was found to be 1.78×10^{-3} atm. Calculate the MW of hemoglobin. ($R = 0.082$ L atm./mole deg.)

$$\pi = \frac{n}{V} RT$$

✓ $T = 25 + 273 = 298$ °K.

✓ $V = 100$ ml = **0.1** L.

$$\pi = \frac{n}{V} RT$$

$$\therefore n = \frac{\pi V}{RT}$$

$$\therefore n = \frac{1.78 \times 10^{-3} \times 0.1}{0.082 \times 298} = 7.28 \times 10^{-6}$$

✓ $n = \text{weight} / \text{MW}$.

✓ $\therefore \text{MW} = \text{weight} / n = 0.5 / 7.28 \times 10^{-6} = 68.68$ g/mol.

Pharmaceutical Application of Colligative Properties

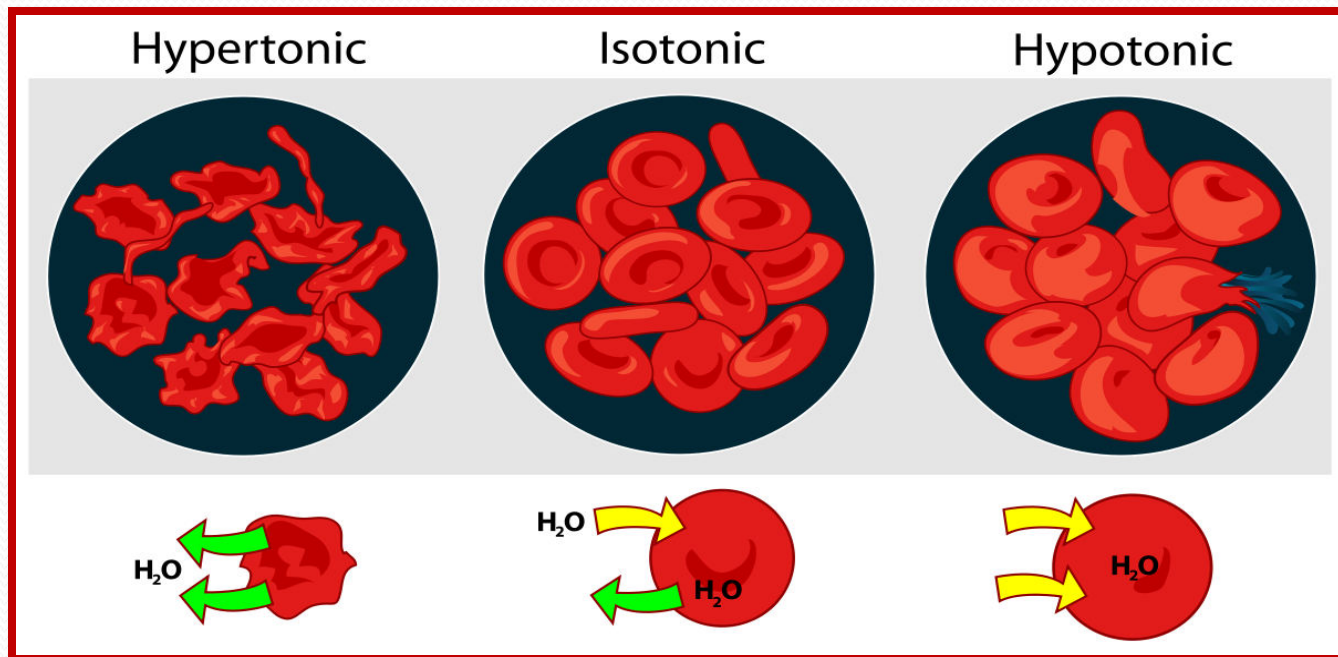
A) Importance of Osmotic Pressure

B) Determination the MW of solutes

C) Determination the Extent of Ionization
(degree of dissociation) of electrolytes

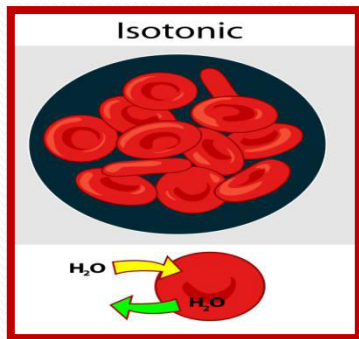
A) Importance of Osmotic Pressure

- ✓ Cell membranes & RBCs membranes are **semipermeable membranes**.
- ✓ According to the O.P. solutions can be classified into:
 - 1) Isotonic solutions.
 - 2) Hypertonic solutions.
 - 3) Hypotonic solutions.



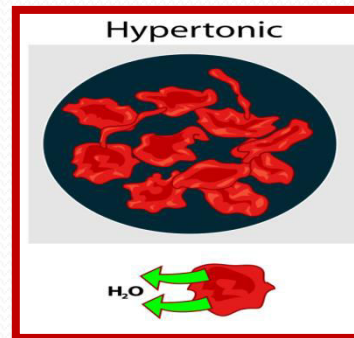
1) Isotonic Solutions

- ✓ The **solution** has the **same salt conc.** of the **blood**.
- ✓ ∴ **Same O.P.** as RBCs & tissue contents.
- ✓ ∴ If injected into blood → RBCs contents are **not disturbed** → **retain their shape & size**.
- ✓ Ex: **0.9% NaCl** aqueous solution.



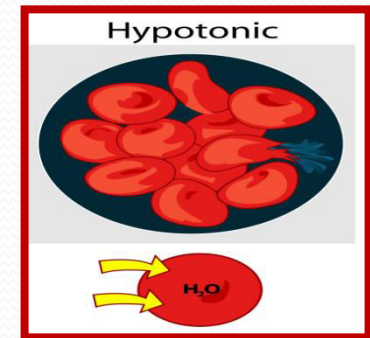
2) Hypertonic Solutions

- ✓ The **solution** has **higher salt conc.** than **RBCs content**.
- ✓ ∴ **Higher O.P.** than RBCs & tissue content.
- ✓ ∴ RBCs contents **move from lower salt conc. (inside RBCs) to higher conc. (outside RBCs)** → **RBCs shrinkage**.
- ✓ Ex: **3% NaCl** solution.



3) Hypotonic Solutions

- ✓ The **solution** has **lower salt conc.** than **RBCs content**.
- ✓ ∴ **Lower O.P.** than RBCs & tissue content.
- ✓ ∴ RBCs contents **move from lower salt conc. (outside RBCs) to higher conc. (inside RBCs)** → **RBCs swelling & rupture (hemolysis)**.
- ✓ Ex: **0.2% NaCl** solution.



The most important pharmaceutical applications of O.P. are:

1) Preparation of Intravenous (IV) & ocular (eye) and nasal solutions:

- **IV & ocular solutions** must be **ISOTONIC** (same O.P. as body fluids).

2) In experimental physiology and transplantation surgery:

- **Tissues should be immersed** in **ISOTONIC salt solutions**.
- This **prevent tissue injuries** that may arise from osmosis.

Isotonicity Adjustment

- ✓ Several methods are used to **calculate the amount of adjusting agent** added to hypotonic solution to become isotonic.

1) Freeze point depression method:

- **F.P. of blood = - 0.52°C.**
- $\therefore$ Dissolved substances in blood & tears **lead to F.P. depression by 0.52°C.**
- $\therefore$ **Any solution has F.P. of - 0.52°C** $\rightarrow$ **Isotonic** with blood & tears.

2) Sodium chloride equivalent method:

- Uses a constant called **NaCl equivalent of drug (E_{NaCl}).**